



Lakshya JEE 2.0 (2024)

Definite Integration

DPP-04

1. $\int_0^{\frac{\pi}{2}} \frac{\sin 8x \log(\cot x)}{\cos 2x} dx =$

- (1) 0 (2) 1
(3) $\frac{1}{2}$ (4) $\frac{\pi}{4}$

2. The value of $\int_{-2\pi}^{5\pi} \cot^{-1}(\tan x) dx$ is equal to

- (1) $\frac{7\pi}{2}$ (2) $\frac{7\pi^2}{2}$
(3) $\frac{3\pi}{2}$ (4) π^2

3. $\int_0^{\pi} \frac{dx}{1+2\sin^2 x}$ is equal to

- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{3\sqrt{3}}$
(3) $\frac{\pi}{\sqrt{3}}$ (4) 0

4. Prove that

$$\int_0^{\frac{\pi}{2}} \log(\sin x) dx = \int_0^{\frac{\pi}{2}} \log(\cos x) dx = -\frac{\pi}{2} \log 2$$

5. $\int_{-\pi/2}^{\pi/2} \frac{|x| dx}{8\cos^2 2x + 1}$ has the value

- (1) $\frac{\pi^2}{6}$ (2) $\frac{\pi^2}{12}$
(3) $\frac{\pi^2}{24}$ (4) $\frac{\pi^2}{18}$

6. $\int_0^{\infty} \frac{\ln x}{1+x^2} dx$

- (1) π (2) 2π
(3) 1 (4) 0

7. $\int_{-\pi/4}^{\pi/4} \log(\cos x + \sin x) dx =$

- (1) $\pi \log 2$ (2) $-\pi \log 2$
(3) $-\frac{\pi}{4} \log 2$ (4) $\pi^2 \log 2$

8. $\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^2 x (\sin x + \cos x) dx =$

- (1) $\frac{2}{15}$ (2) $\frac{4}{15}$
(3) $\frac{6}{15}$ (4) $\frac{8}{15}$

9. If $f(a+b-x) = f(x)$, then $\int_a^b xf(x) dx$ equal to

- (1) $\frac{a+b}{2} \int_a^b f(b-x) dx$
(2) $\frac{a+b}{2} \int_a^b f(x) dx$
(3) $\frac{b-a}{2} \int_a^b f(x) dx$
(4) $\frac{a+b}{2} \int_a^b f(a+b+x) dx$

10. The value of $\int_0^{\pi/2n} \frac{dx}{1+\tan^n(nx)}$ is equal to, ($n \in N$)

- (1) 0 (2) $\frac{\pi}{4n}$
(3) $\frac{\pi}{2n}$ (4) $\frac{\pi}{2}$

11. Given $f(x)f(-x) = 9$ then $\int_{-51}^{51} \frac{dx}{3+f(x)}$ has the value

- equal to
(1) 17 (2) 34
(3) 102 (4) 0

12. The value of $\int_0^{\frac{\pi}{3}} \log(1 + \sqrt{3} \tan x) dx$ is equal to

- (1) $\pi \log 2$ (2) $\frac{\pi}{2} \log 2$
 (3) $\frac{\pi}{3} \log 2$ (4) $\frac{\pi}{4} \log 2$

13. The value of definite integral

$$\int_{-\pi}^{\pi} \frac{x^2}{1 + \sin x + \sqrt{1 + \sin^2 x}} dx,$$

- (1) $\frac{3\pi^2}{2}$ (2) $\frac{\pi^3}{3}$
 (3) $\frac{2\pi^3}{3}$ (4) $\frac{\pi^3}{6}$

14. If $\int_0^{100\pi} \sqrt{1 - \cos 2x} dx$ then the value of I is

- (1) $100\sqrt{2}$ (2) $200\sqrt{2}$
 (3) $50\sqrt{2}$ (4) $25\sqrt{2}$

15. The solution of the integral $\int_0^{2\pi} \sin^4 x \cdot \cos^6 x dx$, is

- (1) $\frac{3\pi}{128}$
 (2) $\frac{\pi}{128}$
 (3) $\frac{5\pi}{128}$
 (4) $\frac{7\pi}{128}$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

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|------------|---------|
| 1. (1) | 9. (2) |
| 2. (2) | 10. (2) |
| 3. (3) | 11. (1) |
| 4. (Proof) | 12. (3) |
| 5. (2) | 13. (2) |
| 6. (4) | 14. (2) |
| 7. (3) | 15. (1) |
| 8. (2) | |



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>